

Deep Learning

Vazgen Mikayelyan

June 12, 2021



- Simple Cross Entropy

$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-y_i^T \log f_w(x_i) \right).$$

Balanced Cross Entropy

- Simple Cross Entropy

$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-y_i^T \log f_w(x_i) \right).$$

- Balanced Cross Entropy

$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-(\alpha * y_i)^T \log f_w(x_i) \right).$$

Where $\alpha^T = [\alpha_1, \dots, \alpha_k]$, $\alpha_i \in (0, 1)$ and $\sum_{i=1}^k \alpha_i = 1$.

1 Basic Recurrent Neural Networks

Sequential Data

- What is sequential data?

Sequential Data

- What is sequential data?
 - Time series

Sequential Data

- What is sequential data?
 - Time series
 - Text

Sequential Data

- What is sequential data?
 - Time series
 - Text
 - Speech

Sequential Data

- What is sequential data?
 - Time series
 - Text
 - Speech
 - Audio

- What is sequential data?
 - Time series
 - Text
 - Speech
 - Audio
 - Video

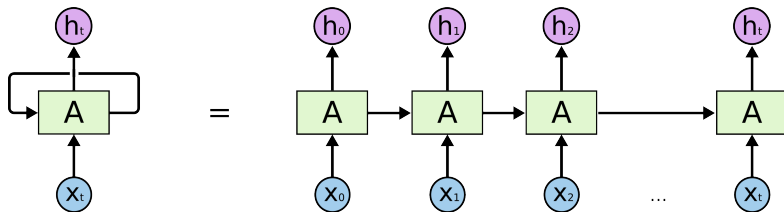
Sequential Data

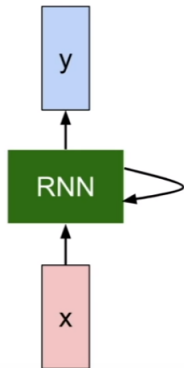
- What is sequential data?
 - Time series
 - Text
 - Speech
 - Audio
 - Video
- Why don't to use known networks?

- What is sequential data?
 - Time series
 - Text
 - Speech
 - Audio
 - Video
- Why don't to use known networks?
 - Inputs, outputs can be different lengths in different examples.

- What is sequential data?
 - Time series
 - Text
 - Speech
 - Audio
 - Video
- Why don't to use known networks?
 - Inputs, outputs can be different lengths in different examples.
 - Doesn't share features learned across different positions of sequence.

Basic RNNs





$$h_t = f_W(h_{t-1}, x_t)$$



$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$$

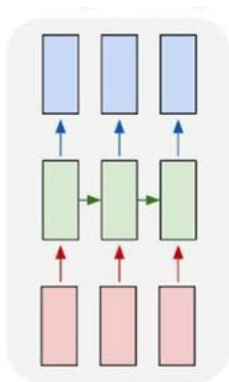
$$y_t = W_{hy}h_t$$

Different Types of RNNs

- Many to many

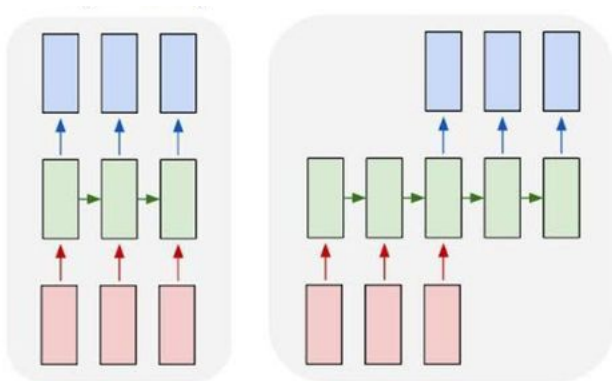
Different Types of RNNs

- Many to many



Different Types of RNNs

- Many to many

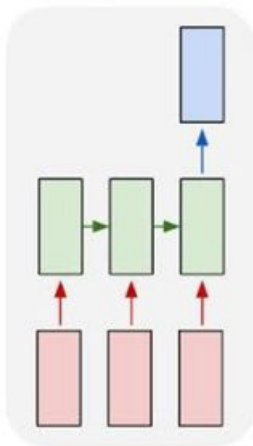


Different Types of RNNs

- Many to one

Different Types of RNNs

- Many to one

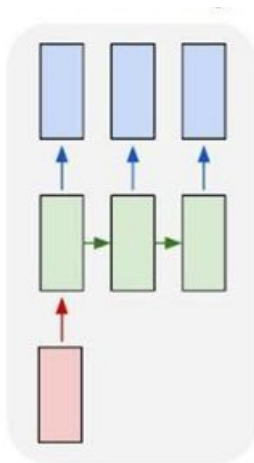


Different Types of RNNs

- One to many

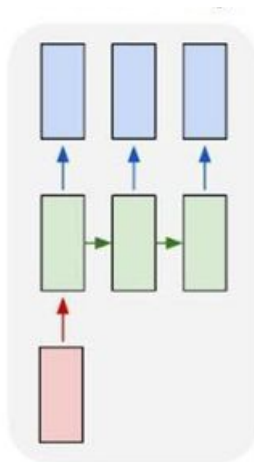
Different Types of RNNs

- One to many



Different Types of RNNs

- One to many



- One to one?

Problem of Long Term Dependencies

